

ECON211: Economic Statistics – Midterm

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- Total points: 64
- Time: 50 minutes
- Instructions: Write final answers near each question; circle the answers clearly.

Emergency Response Times

A city's emergency medical services department analyzes its response times. They find that the time T (in hours) it takes for the first respondents to arrive after a call follows a distribution with probability density function given below:

$$f_T(t) = \begin{cases} 1 - 0.5t, & 0 \leq t \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

A response is officially considered "on time" if the responders arrive within the first 20 minutes.

1. (3 points) Find the probability that a single call is answered "on time."
2. (4 points) Let $F_T(t)$ be the CDF of the response time. Write its functional form for $0 \leq t \leq 2$.
3. (4 points) Calculate the **median** response time m .
4. (4 points) Find the expected response time $E[T]$.
5. (5 points) Find the variance of the response time, $Var(T)$.
6. (4 points) For 100 independent calls, let X be the number of calls answered on time. Identify by name the distribution of X and its parameters.
7. (4 points) For 10 calls, write the expression for the probability that **at least two** calls are **not** answered on time.

Miscellaneous problems

1. (4 points) In a group of 100 people, 70 are employed, 50 have a university degree, and 30 have both. If a person is chosen at random and they have a degree, what is the probability they are employed?
2. An urn has 1 white and 4 black balls. You take out 2 balls without replacement. Let X be the number of white balls among the ones you take out.
 - (2 points) Identify the name of the distribution of X .
 - (2 points) What is the probability that $X = 1$?
 - (2 points) Find $E[X]$.
3. (4 points) You bet \$1 on an integer from 1 to 10. If a random integer matches yours, you win \$9 (net); else, you lose \$1. Find $Var(\text{Winnings})$.
4. (5 points) Laboratory mice develop a specific condition each month with probability p . If, on average, mice develop the condition at the end of the 5th month, find the value of p .
5. (6 points) On average, how many times do you need to roll a fair six-sided die before all six different numbers have showed up at least once?